AI101 MINI PROJECT • GROUP 74

Loan Eligibility Prediction

Predicting whether a loan application is likely to be approved using a Logistic Regression classification model.

TEAM MEMBERS

Tanush Metla • Neil Patil • Abhinav Karoli • Devnandan Menon

GitHub: https://github.com/2026aib1062-creator/loan-eligibility-prediction-project-grp74.git

AI101 Mini-Project

Problem & dataset

What the project predicts and the data used

THE PROBLEM

To predict loan approval eligibility based on applicant financial and demographic profile

The model learns patterns from previous loan applications and classifies an application as approved (Y) or not approved (N) using logistic regression

Goal: support a consistent, data-driven first-pass assessment.

DATASET SNAPSHOT

614 original records	**13** columns
502 rows after cleaning	**11** model input features

Source: Kaggle – Eligibility Prediction for Loan
https://www.kaggle.com/datasets/devzohaib/eligibility-prediction-for-loan

Approach & model pipeline

From raw application data to a loan-status prediction

01

CLEAN DATA

Drop ID/Gender; remove rows missing selected fields; convert Dependents

02

PREPARE FEATURES

One-hot encode categorical columns using the get_dummies() function; separate X and y

03

SPLIT DATA

Stratified split: 67% train / 33% test (random_state = 100)

04

SCALE + TRAIN

StandardScaler; Logistic Regression (max_iter = 100)

05

PREDICT

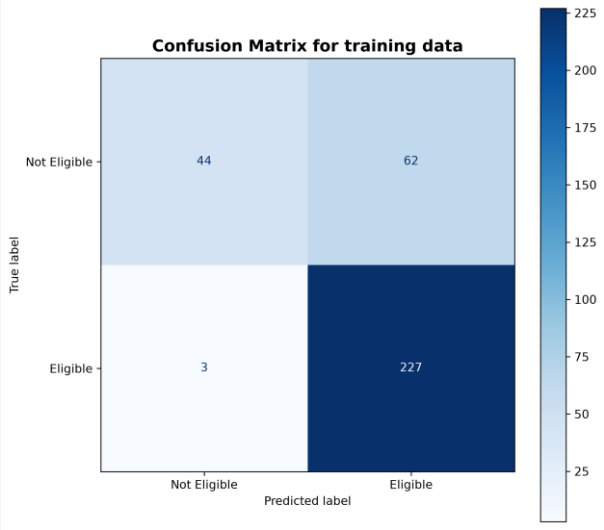
Predict test labels; calculate precision, recall and F1-score

LIBRARIES Pandas • NumPy • Matplotlib • Scikit-learn

Output: predicted loan status (Y / N) for each record in the test set.

Results

The Confusion Matrix of the model performance on testing data



The Model Performance Metrics on the training data

Precision: 0.785
Recall: 0.987
F1-score: 0.875

The Model Performance Metrics on the testing data

Precision: 0.807
Recall: 0.965
F1-score: 0.879

Screenshot of the output:

```
C:\Users\devme\OneDrive\Desktop\Progamming\loan-eligibility-prediction-project-grp74>python main.py
Data loaded and cleaned.
Shape of clean dataset: (502, 12)
Logistic Regression model trained successfully!

Made predictions on test data. The following are the performance metrics for the trained logistic regression model.

Precision: 0.807
Recall: 0.965
F1-score: 0.879

Made predictions on training data. The following are the performance metrics for the trained logistic regression model.

Precision: 0.785
Recall: 0.987
F1-score: 0.875

Saved prediction comparison; 'Output/Loan Eligibility prediction comparison.png'
Saved confusion matrix; 'Output/confusion_matrix_for_training_data.png'
Saved confusion matrix; 'Output/confusion_matrix_for_testing_data.png'
```

Learnings & team contributions

What the team practiced while building the classifier

WHAT WE LEARNED

- Data cleaning is essential before model training.
- Categorical variables must be encoded for the model.
- A stratified train/test split helps preserve class proportions.
- Scaling and Logistic Regression form the core modelling pipeline.
- Precision, recall and F1-score provide different views of model performance.

WHO DID WHAT

Tanush Metla

PPT lead; assigned comparison/bar-graph code

Neil Patil

Documentation lead; assigned data cleaning and model training/prediction. Created the README.md file

Abhinav Karoli

Video lead; assigned train/test split and model analysis

Devnandan Menon

Execution lead; Created the requirements.txt file. Plotted the confusion matrix. Cleared the bugs in the program

AI-use disclosure

AI was used for graphing the performance metrics,to create organised slide and taken reference for making REAME.md file